

RKT2

Linear Algebra

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An engineer uses software with 16-digit precision to row-reduce the augmented matrix of a 3×3 system for a bridge sensor network. The result is $[1 \ 0 \ 2 \ | \ 4]$, $[0 \ 1 \ -1 \ | \ 1]$, $[0 \ 0 \ 0 \ | \ 7]$. The project is funded for two years. How many solutions does the system have?

- A. Exactly one.
- B. It depends on rounding in the software.
- C. Infinitely many, because there is a zero row and a free variable.
- D. None; the system is inconsistent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A graphics programmer on a team of five stores two transforms as $A = [[1, 2], [3, 4]]$ and $B = [[0, 1], [1, 0]]$. Her code runs at 60 frames per second. What is AB ?

- A. $[[3, 4], [1, 2]]$
- B. $[[0, 2], [3, 0]]$
- C. $[[2, 1], [4, 3]]$
- D. $[[1, 3], [2, 4]]$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A game studio's sprite is 64 pixels wide. An artist wants to rotate it 90° counterclockwise about the origin. The studio released its last title in April. Which standard matrix does this?

- A. $[[0, 1], [-1, 0]]$
- B. $[[1, 1], [0, 1]]$
- C. $[[0, 1], [1, 0]]$
- D. $[[0, -1], [1, 0]]$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

What does the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ do to points in the plane?

- A. Reflects them across the y-axis
- B. Applies a vertical shear
- C. Reflects them across the x-axis
- D. Rotates them 180°

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A landscape designer who charges \$95 an hour maps a garden bed with area 6 square meters through the linear map with matrix $\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$. The client wants the plan by June. What is the area of the image?

- A. 10 square meters
- B. 42 square meters
- C. 84 square meters
- D. 60 square meters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A linear map with determinant -3 sends a region to an image with area 45. What was the original area?

- A. 15
- B. 135
- C. -15
- D. 45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 3×3 matrix A in an economics model has $\det A = 4$. The model uses data from 12 industries. An analyst doubles every entry. What is $\det(2A)$?

- A. 8
- B. 32
- C. 16
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A 3×3 matrix A has $\det A = 2$, and $\det(kA) = 54$. What is k ?

- A. 27
- B. 3
- C. 18
- D. About 5.2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A laptop computes $\det A = 0$ for a 3×3 matrix A in 0.002 seconds. The matrix comes from a traffic model with three roads. Which statement must be true?

- A. $Ax = b$ has a unique solution for every b .
- B. $Ax = 0$ has a nontrivial solution.
- C. The columns of A are linearly independent.
- D. A is the zero matrix.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A student in a summer bridge program of 18 people must invert $A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$ by hand. Which matrix is A^{-1} ?

- A. $(1/10) \begin{bmatrix} 6 & 7 \\ 2 & 4 \end{bmatrix}$
- B. $10 \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$
- C. $(1/10) \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$
- D. $(1/10) \begin{bmatrix} 4 & -7 \\ -2 & 6 \end{bmatrix}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A data scientist with a large research grant looks at the set W of all vectors (x, y, z) in $\mathbb{R}^3$ with $x + y + z = 1$. Her lab has four servers. Is W a subspace of $\mathbb{R}^3$?

- A. No; W does not contain the zero vector.
- B. Yes; W is closed under addition.
- C. No; the only subspace of $\mathbb{R}^3$ is $\mathbb{R}^3$ itself.
- D. Yes; W is a plane in $\mathbb{R}^3$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which of these is a subspace of $\mathbb{R}^2$?

- A.** All (x, y) with $y = 3x$
- B.** All (x, y) with $xy = 0$
- C.** All (x, y) with $x \geq 0$ and $y \geq 0$
- D.** All (x, y) with $y = 3x + 1$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An engineer uses software ~~with 16-digit precision~~ to row-reduce the augmented matrix of a 3×3 system ~~for a bridge sensor network~~. The result is $[1 \ 0 \ 2 \ | \ 4]$, $[0 \ 1 \ -1 \ | \ 1]$, $[0 \ 0 \ 0 \ | \ 7]$. ~~The project is funded for two years~~. How many solutions does the system have?

Target word: number of solutions **Decides:** The last row says $0 = 7$.

A	Dropped element	Drops the third row and ignores that x_3 has no pivot.
B	Camouflage	$0 = 7$ is exact; precision does not change it.
C	Confusion set	A zero row gives free variables only when the right side is also 0.
D	Correct	A row $[0 \ 0 \ 0 \ \ 7]$ is a contradiction.

2. Answer C Concept 2

A graphics programmer ~~on a team of five~~ stores two transforms as $A = [[1, 2], [3, 4]]$ and $B = [[0, 1], [1, 0]]$. ~~Her code runs at 60 frames per second~~. What is AB ?

Target word: the product AB **Decides:** Rows of A dotted with columns of B .

A	Inverted operation	That is BA , which swaps A 's rows.
B	Confusion set	Multiplies entry by entry.
C	Correct	Row 1: $(1 \cdot 0 + 2 \cdot 1, 1 \cdot 1 + 2 \cdot 0) = (2, 1)$; row 2: $(4, 3)$. B swaps A 's columns.
D	Adjacent category	That is A^T , the transpose.

3. Answer D Concept 3

A game studio's sprite is ~~64 pixels wide~~. An artist wants to rotate it 90° counterclockwise about the origin. ~~The studio released its last title in April~~. Which standard matrix does this?

Target word: rotation matrix **Decides:** $e_1 \rightarrow (0, 1)$ and $e_2 \rightarrow (-1, 0)$ become the columns.

A	Inverted operation	That rotates 90° clockwise.
B	Adjacent category	That is a horizontal shear.
C	Confusion set	That reflects across the line $y = x$.
D	Correct	Columns are the images $(0, 1)$ and $(-1, 0)$.

4. Answer C Concept 3 · transfer

What does the matrix $[[1, 0], [0, -1]]$ do to points in the plane?

Target word: identify the transformation **Decides:** $(x, y) \rightarrow (x, -y)$.

A	Inverted operation	That would be $[[-1, 0], [0, 1]]$.
B	Adjacent category	A shear has a nonzero off-diagonal entry.
C	Correct	x is kept and y changes sign.
D	Confusion set	That would be $[[-1, 0], [0, -1]]$.

5. Answer D Concept 4

A landscape designer ~~who charges \$95 an hour~~ maps a garden bed with area 6 square meters through the linear map with matrix $\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$. ~~The client wants the plan by June.~~ What is the area of the image?

Target word: image area **Decides:** $\det = 3 \cdot 4 - 1 \cdot 2 = 10$.

A	Dropped element	That is the scale factor; the original area is dropped.
B	Confusion set	Uses the trace, $3 + 4 = 7$.
C	Inverted operation	Adds $12 + 2$ instead of subtracting.
D	Correct	$6 \times 10 = 60$.

6. Answer A Concept 4 • transfer

A linear map with determinant -3 sends a region to an image with area 45. What was the original area?

Target word: solve for the original area **Decides:** Image area = $|\det| \times$ original area.

A	Correct	$45 / 3 = 15$.
B	Inverted operation	Multiplies instead of dividing.
C	Adjacent category	Keeps the sign; the sign shows orientation, and area uses $ \det $.
D	Wrong stem number	That is the image area.

7. Answer B Concept 5

A 3×3 matrix A in an economics model has $\det A = 4$. ~~The model uses data from 12 industries.~~ An analyst doubles every entry. What is $\det(2A)$?

Target word: determinant of a scalar multiple **Decides:** $\det(kA) = k^n \det A$ with $n = 3$.

A	Dropped element	2×4 drops the power n .
B	Correct	$2^3 \times 4 = 32$.
C	Confusion set	$2^2 \times 4$ uses the 2×2 rule.
D	Inverted operation	Adds 2 instead of multiplying.

8. Answer B Concept 5 • transfer

A 3×3 matrix A has $\det A = 2$, and $\det(kA) = 54$. What is k ?

Target word: solve for the scalar **Decides:** $k^3 \times 2 = 54$, so $k^3 = 27$.

A	Dropped element	That is k^3 ; the cube root is dropped.
B	Correct	The cube root of 27 is 3.
C	Inverted operation	Divides 54 by 3 instead of taking a cube root.
D	Confusion set	$\sqrt{27}$ uses $n = 2$.

9. Answer B Concept 6

A laptop computes $\det A = 0$ for a 3×3 matrix A in 0.002 seconds. The matrix comes from a traffic model with three roads. Which statement must be true?

Target word: consequence of $\det = 0$ **Decides:** $\det A = 0$ means A is not invertible.

A	Confusion set	That is a property of invertible matrices.
B	Correct	A singular matrix has dependent columns.
C	Inverted operation	That holds only when $\det A \neq 0$.
D	Absolute language	Many nonzero matrices have determinant 0.

10. Answer C Concept 7

A student in a summer bridge program of 18 people must invert $A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$ by hand. Which matrix is A^{-1} ?

Target word: 2×2 inverse **Decides:** $\det = 24 - 14 = 10$; swap a and d , negate b and c .

A	Dropped element	Drops the sign changes on b and c .
B	Inverted operation	Multiplies by the determinant instead of dividing.
C	Correct	Applies the 2×2 formula with $\det 10$.
D	Confusion set	Negates b and c but does not swap a and d .

11. Answer A Concept 8

A data scientist with a large research grant looks at the set W of all vectors (x, y, z) in $\mathbb{R}^3$ with $x + y + z = 1$. Her lab has four servers. Is W a subspace of $\mathbb{R}^3$?

Target word: subspace test **Decides:** $(0, 0, 0)$ gives $0 \neq 1$.

A	Correct	Every subspace must contain 0.
B	Confusion set	It is not: two vectors in W add to a sum of 2.
C	Absolute language	Lines and planes through the origin, and $\{0\}$, are subspaces too.
D	Adjacent category	Only planes through the origin are subspaces.

12. Answer A Concept 8 · transfer

Which of these is a subspace of $\mathbb{R}^2$?

Target word: pick the subspace **Decides:** Contains 0 and is closed under both operations.

A	Correct	A line through the origin passes all three tests.
B	Adjacent category	$(1, 0) + (0, 1) = (1, 1)$ leaves the set; not closed under addition.
C	Confusion set	Not closed under scalar multiplication by -1 .
D	Dropped element	Misses the origin.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Consistency from row reduction	A system $Ax = b$ is inconsistent when row reduction gives a row $[0 \ 0 \ \dots \ 0 \ \ c]$ with $c \neq 0$; free variables matter only if the system is consistent.	Reading any zero row on the left as “infinitely many solutions.”	An echelon form with a zero row on the coefficient side.
2	Matrix multiplication is not commutative	Entry (i, j) of AB is row i of A dotted with column j of B , and in general $AB \neq BA$.	Multiplying entry by entry, or computing BA when AB is asked.	Two named matrices and a product in a specific order.
3	Standard matrices of transformations	The columns of the standard matrix are the images of e_1 and e_2 ; rotation by 90° counterclockwise sends e_1 to $(0, 1)$ and e_2 to $(-1, 0)$.	Confusing clockwise with counterclockwise, or a rotation with a reflection.	Rotate, reflect, shear, or dilate in the plane.
4	Determinant as an area scale factor	A linear map multiplies every area by $ \det A $.	Using the trace, or keeping a negative sign for area.	A region’s area before or after a transformation.
5	Determinant of a scalar multiple	For an $n \times n$ matrix, $\det(kA) = k^n \det A$.	Writing $\det(kA) = k \det A$.	A scalar times a matrix inside det.
6	Invertible matrix theorem	For square A , $\det A \neq 0 \Leftrightarrow A$ is invertible $\Leftrightarrow$ the columns are independent $\Leftrightarrow Ax = 0$ has only the trivial solution.	Mixing the conditions for invertible and singular matrices.	“ $\det A = 0$ ” and a list of claims.
7	Inverse of a 2×2 matrix	For $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is $(1 / (ad - bc)) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.	Forgetting to swap a and d or to negate b and c , or multiplying by the determinant.	“Find A^{-1} ” for a 2×2 matrix.
8	Subspace test	A subset is a subspace only if it contains the zero vector and is closed under addition and scalar multiplication.	Calling any line or plane a subspace, even one that misses the origin.	A set defined by an equation or inequality.

Course scope (public description): Linear Algebra is the study of the algebra of curve-free functions extended into three-or-higher-dimensional space. It covers the knowledge and skills necessary to apply vectors, matrices, matrix theorems, and linear transformations and to use appropriate technology to model and solve real-life problems. It also covers properties of and proofs about vector spaces. Topics include linear equations and their matrix-vector representation $Ax=b$, row reduction, linear transformations and their matrix representations (shear, dilation, rotation, reflection), matrix operations, matrix inverses and invertible matrix characterizations, computing determinants, relating determinants to area and volume, and axiomatic and intuitive definitions of vector spaces and subspaces and how to prove theorems about them. College Geometry and Calculus II are prerequisites for this course.